

Copenhagen Airport (Københavns Lufthavn) · flow reference

Field	Value
Site ID	cph
Label	Copenhagen Airport / Københavns Lufthavn
Code	CPH (IATA) / EKCH (ICAO)
Tenant	op-cph-airports (CPH Airports A/S)
Onboarded	2026-08 (seed)
Site type	airport
Operating mode	live (sim overlay for training scenarios)

Reference site for the per-site flow doc pattern. Every other site follows this shape.

1. Overview

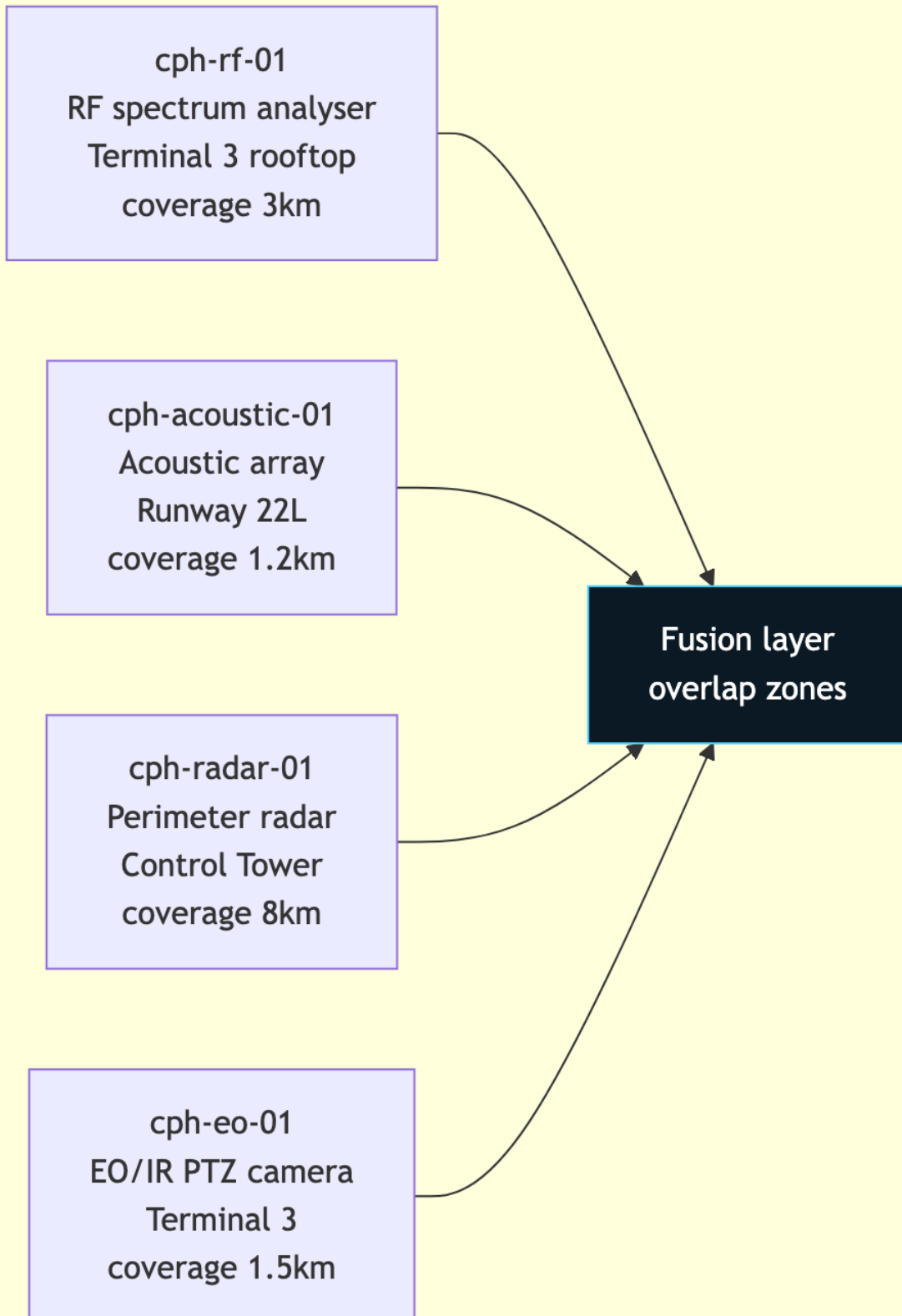
Copenhagen Airport is Denmark's primary international airport, ~30 million annual passengers, 22 airlines, hub for SAS. The ISR C2 platform watches airspace surrounding the airport perimeter for drone detections that could disrupt takeoff / landing operations. Detections trigger a cascade to Trafikstyrelsen (airspace regulator, NOTAM authority), Politi København (local police + AKS specialists), and — for hostile signatures — Forsvarets Efterretningstjeneste (FE) plus Politiets Efterretningstjeneste (PET) for pattern-of-life logging.

Normal operations: ~700 movements/day (takeoff + landing combined). Class D controlled airspace up to 4500 ft AGL. Base coordination with ATC on 118.100 MHz. Runway 04L/22R is primary; 04R/22L is secondary and cross-wind alternate.

Sensor deployment: RF spectrum analyser on Terminal 3 rooftop, acoustic array near Runway 22L, perimeter radar on Control Tower, EO/IR PTZ camera on Terminal 3. Multi-modality fusion at the platform layer resolves the "same drone, multiple detections" case + cross-references cooperative-traffic (Naviar) to explain away legitimate GA traffic.

2. Sensor layout

Copenhagen Airport site fabric



Sensor ID	Modality	Position	Coverage	Notes
cph-rf-01	RF spectrum	55.6181, 12.6560, alt 15m	3km radius	Full broadband 400 MHz - 6 GHz. Detects DJI OcuSync + FPV + Shahed autonomous silence signature
cph-acoustic-01	Acoustic	55.6175, 12.6540, alt 8m	1.2km radius	32-mic array. Distinguishes rotary from turbojet from electric. Weather-degraded above 20 knots wind
cph-radar-01	Radar (X-band)	55.6190, 12.6600, alt 45m	8km radius	Primary track source. Rain-degraded above 15mm/hr. Small-UAV detection down to 0.01 m² RCS
cph-eo-01	EO/IR	55.6183, 12.6555, alt 25m	1.5km radius	Slaves to radar track for auto-cue. Manual override via Mission Console. Thermal imaging for night ops

Blind spots: the western perimeter (~500m sector centred on 55.616, 12.630) has degraded RF coverage due to terminal building shadowing. Radar + acoustic compensate. EO cannot resolve targets in that sector — a detection there always renders without visual confirmation and is flagged in the case-file emphasis.

Sensor id note: the four sensor ids above (`cph-rf-01` , `cph-acoustic-01` , `cph-radar-01` , `cph-eo-01`) are illustrative modality archetypes for narrative clarity. The actual seed data in `src/sites.js` registers CPH with 22 online / 24 total multi-modality Radxa nodes (ids `N01` - `N24` , each combining RF + acoustic + visual on one board, with the HackRF SDR core on N08). When the site definition schema per `docs/integration-contracts.md` Section 3 migrates the seed data, each Radxa node becomes one manifest `sensors[]` entry with multi-modality flags — the four-archetype narrative here still holds as a role-grouping.

Overlap zones: central runway area (~200m radius around 55.618, 12.656) has full 4-modality coverage. Detections here get the highest fusion confidence.

3. Domain scope + operating mode

- **Domain scope:** `[aviation, ground]`
- **Operating mode:** live (all four sensors real). Sim mode overlay available for training scenarios — enabled per operator session, marked with `source: 'sim'` on every detection.
- **Airspace regulator:** Trafikstyrelsen. NOTAM authority for restriction issuance.

4. Default cascade recipients

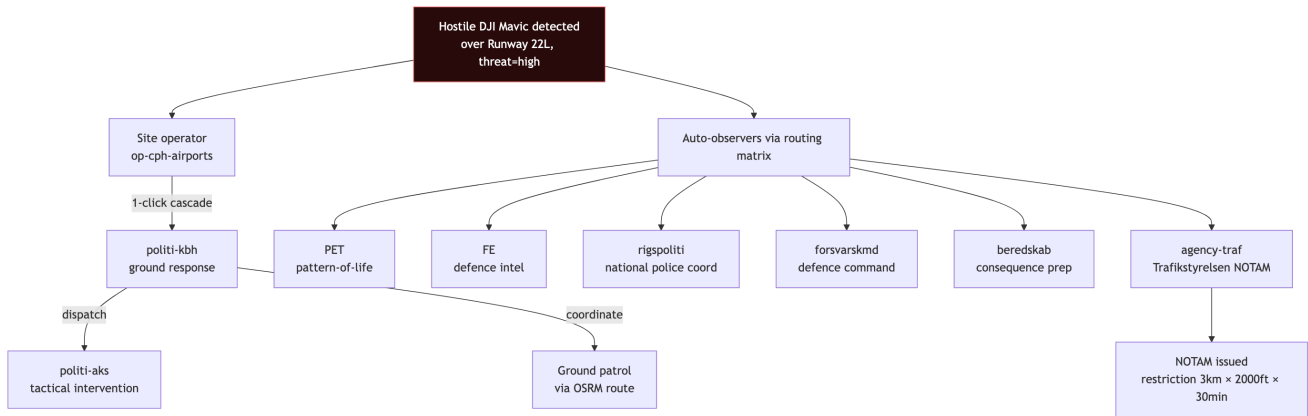
Event shape	Recipients	Rationale
Any hostile	pet, fe, politi-kbh	Intel + local ground authority baseline
Hostile + high threat	+ forsvarskmd, rigspoliti, beredskab, agency-traf	National command + emergency response + NOTAM
Cruise / missile signature	+ flv-karup (Air Force), flv-skrydstrup (F-35 QRA)	Airspace intercept
FPV / kamikaze	+ politi-aks (Aktionsstyrken)	Tactical intervention specialist
Drone swarm	+ full national command chain + beredskab	Multi-threat escalation
Cyber signature	+ rigspoliti-nc3, cert-dkcert	Forensic + cyber attribution

Mass gathering event onsite (e.g. VIP flight, terminal evacuation)	+ kbr-taarndby, region-hst	Fire/rescue + medical
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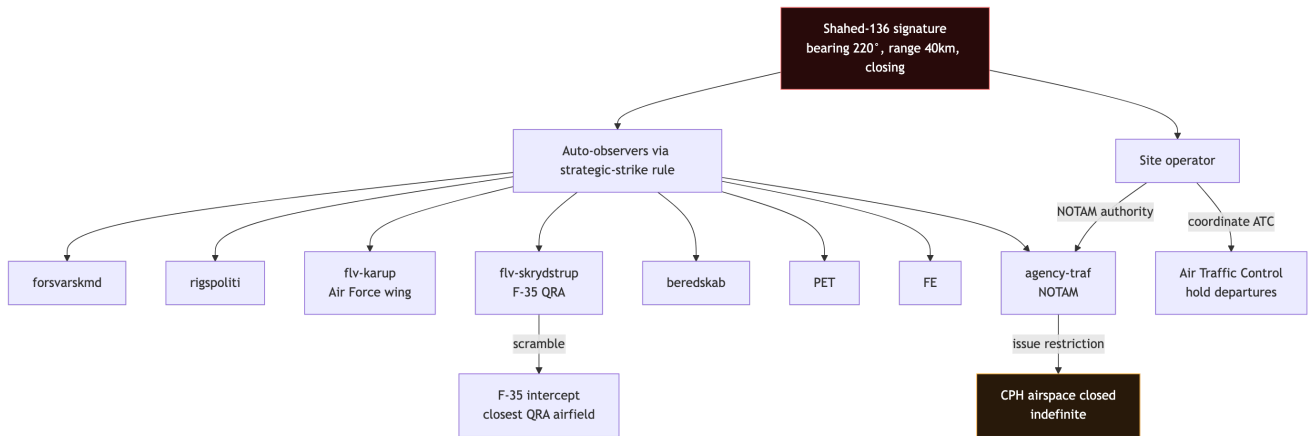
Always-observers (baseline, regardless of routing rule): agency-traffic (Trafikstyrelsen is baseline for aviation domain).

5. Cascade tree for common event shapes

Hostile quadcopter over Runway 22L

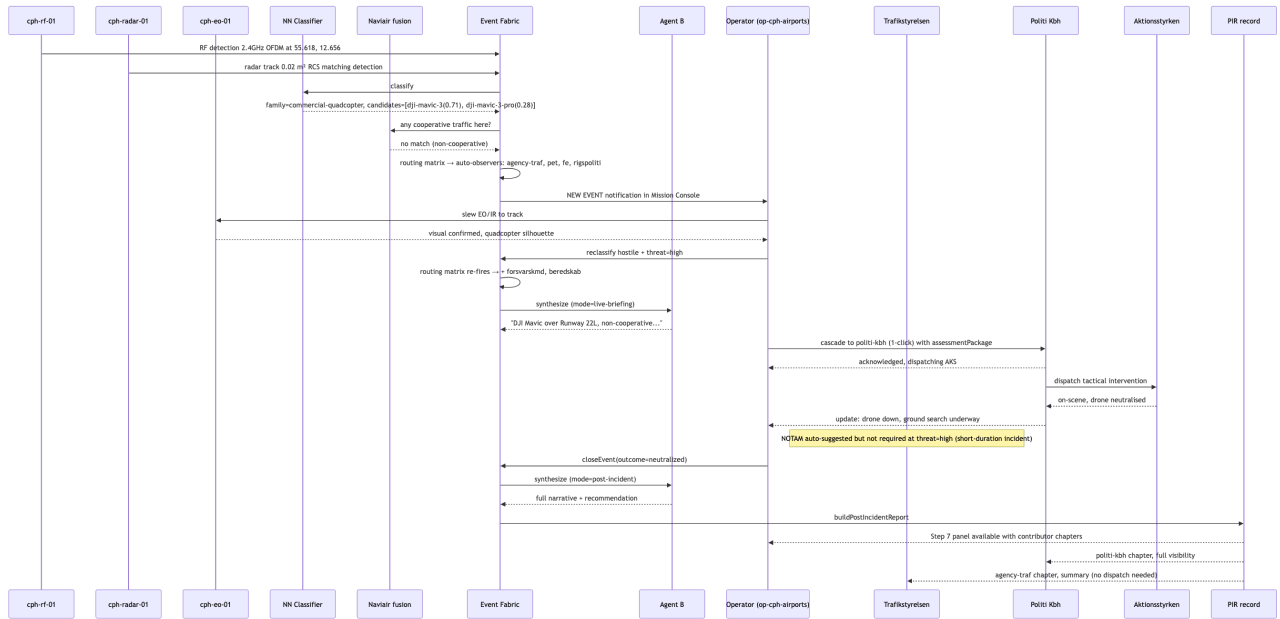


Shahed signature approaching CPH airspace



6. Full flow: detection → closed report

Canonical case: hostile DJI Mavic detected over Runway 22L, operator escalates, Politi Kbh dispatches Aktionsstyrken, drone neutralised, event closed.



7. Edge cases

- **Runway proximity:** any detection within 500m of an active runway triggers auto-suggestion for NOTAM issuance regardless of classification. Operator can dismiss but the suggestion always appears.
- **Cross-boundary events:** drones crossing FROM CPH airspace INTO Roskilde (EKRK) airspace 15km southwest — the platform links events via `LinkedEventIds` (Phase 7 xlink graph). Both sites' operators see the linked event in their Mission Console chain view.
- **VIP flight movements:** when a VIP flight is scheduled (STA/STD windows published in Naviar), the platform temporarily elevates baseline threat to "high" for detections within 5km. Reverts after VIP movement completes.
- **False positives from Kastrup marina:** occasional wave-clutter radar returns from the marina 2km east. Cooperative-traffic fusion should filter, but if it doesn't, operator marks as false-positive + feedback log records for classifier tuning.
- **Weather-restricted operations:** during heavy fog (<800m RVR) or precipitation >15mm/hr, radar coverage degrades. Case-file flags "sensor degraded" + confidence downgraded accordingly. Cascade recipients still notified; operator makes the call whether to hold based on their SOP.
- **Flocks near coastal approach:** bird flocks over Kastrup can produce non-cooperative multi-track patterns that resemble drone swarm. Cooperative-traffic fusion doesn't cover birds; NN classifier has bird-vs-drone distinguisher that flags with `target_class_hint: bird` from the radar layer.
- **Airshow days (annual):** cooperative traffic layer receives an airshow-mode flag from Trafikstyrelsen — expected non-scheduled traffic patterns are pre-annotated, reduces false-positive rate for the day.

8. Escalation SLA overrides

Event shape	Platform default	CPH override
hostile + high	5 min	3 min (aviation-critical)
cruise / missile	2 min	2 min (no override)
hostile + medium	10 min	5 min
commercial quadcopter, hostile + low	15 min	10 min

Reasoning: CPH is Denmark's primary hub; downstream cascade delays translate directly to airline schedule impact + passenger delays. Tight SLAs reflect the operational cost of slow escalation.

9. Regulatory context

Class D controlled airspace up to 4500 ft AGL over Copenhagen Airport, transitioning to Class C above. Coordination with Air Traffic Control on 118.100 MHz (Tower), 121.850 MHz (Ground), 119.775 MHz (Radar).

NOTAM authority: Trafikstyrelsen (Danish Civil Aviation Authority). NOTAM issuance for drone incursions:

- Restriction shape: cylinder centred on detection, radius per event scale, altitude 0 to 4500 ft AGL, duration per event scale
- Timing: NOTAM issued within 3 min of Trafikstyrelsen cascade acknowledgement for high-threat events
- Coordination handshake: Trafikstyrelsen operator confirms restriction shape with CPH operator before publishing

Reporting obligations:

- Every event with runway proximity (<500m) or altitude within active flight levels: incident report to Trafikstyrelsen within 24 hours
- Every event with airline schedule impact: report to affected airline within 2 hours
- Every hostile-classified event: report to PET within 24 hours (regardless of whether they were auto-cascaded)

Data retention:

- Aviation Safety Reporting: 10 years (statutory)
- Personal data on detected operators (if identified): GDPR-compliant, 6 months default retention with legal-hold override
- Sensor raw data: 90 days rolling in Azure Blob (cheaper tier after 30 days)

Customer contract compliance:

- SLA: 99.5% platform uptime during operational hours (05:30 - 24:00 CET)
- Reporting cadence: monthly incident summary to CPH Airports A/S ops team
- Sensor maintenance windows: coordinated with airport operations (avoid peak movement periods)

10. Contacts

Role	Name	Contact	On-call schedule
CPH Airports Ops Lead			24/7 rotating
CPH Airports Escalation			24/7 duty officer
Trafikstyrelsen liaison			Business hours + on-call
Politi Kbh drone unit		114 (dispatch)	24/7
ISR technical integration			Business hours

11. Change history

Date	Change	Author
2026-08	Initial site onboarding as seed data	ISR C2 build
2026-09-13	Flow reference doc written per new per-site pattern	ISR C2 build

Reference: docs/integration-contracts.md Section 3 "Site definition" for the machine-readable manifest schema this doc accompanies.